MIL-B-8948A
30 November 1972
SUPERSEDING
MIL-B-8948(ASG)
25 JULY 1967

MILITARY SPECIFICATION

BEARINGS, PLAIN, ROD END, TFE LINED, SELF-ALIGNING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers plain rod end bearings which are self-aligning and self-lubricating by incorporating tetrafluoroethylene (TFE) in a liner between the inner ring (ball) and outer ring (race) for use in the temperature range -65° F to +250° F.

1.2 Classification - Bearings shall be of the following classes, as specified (see 6.2):

Class 1 Externally threaded shank
Class 2 Internally threaded shank

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-B-197	Bearings, Anti-Friction, Associated Parts and Subassemblies, Packaging of
MIL-I-6868	Inspection Process, Magnetic Particle

FSC 3120

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MIL-S-8879	Screw Threads, Controlled Radius Root with Increased Minor Diameter; General Specification for
MIL-B-7838	Bolt, Internal Wrenching, 160 KSI FTU
MIL-B-8942	Bearings, Plain, TFE Lined, Self-Aligning
MIL-D-1000	Drawing, Engineering, and Associated List

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MS21230	Bearing; Plain, Self-Aligning, Grooved Outer Ring, TFE Lined, Wide
MS21242	Bearing, Plain, Rod End, Self-Aligning, TFE Lined, Externally Threaded
MS21243	Bearing, Plain, Rod End, Self-Aligning, TFE Lined, Internally Threaded

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute

ANSI B46.1 - Surface Texture (Surface Roughness, Waviness and Lay)

Copies of the above publication may be obtained from the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicage, Ill. 60606.)

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3. REQUIREMENTS

3.1 Qualification - The bearings furnished under this specification shall be a product which is in accordance with the applicable Military Standard (MS), and contains a bearing cartridge in accordance with MS21230 that has been qualified to MIL-B-8942, and which has been subjected to and which has passed the qualification tests specified herein, and which has been listed on or approved for listing on the applicable Qualified Products List.

3.1.1 Product design change - Any change in product design or description will require requalification of the product to an extent determined by the activity responsible for qualification.

3.2 Material - The ball and race shall be in accordance with MS21230. The rod end body materials shall be in accordance with MS21242 or MS21243. The liner shall include TFE in such manner that it will conform to all the requirements of MIL-B-8942 and this specification.

3.2.1 Plating - Cadmium plating shall be in accordance with QQ-P-416, type I, class 2.

3.3 Design - Bearing design shall conform to MS21242 and MS21243.

3.4 Construction - Construction shall be such that all relative motion will be between the liner and the ball. The bearing shall not have loading slots. Except as otherwise specified, the details of the working parts shall be optional.

3.4.1 Dimensions and tolerances - Dimensions and tolerances shall be as specified on the applicable MS. Dimensions not shown shall be at the option of the manufacturer.

3.4.2 Surface texture - Surface finishes shall be in accordance with the applicable MS and ANSI B46.1. Bearings shall be free of any defects which may be detrimental to satisfactory installation, performance or bearing life, as defined by this specification.

3.4.3 Lubrication - Initial lubrication by the manufacturer will be permitted. Relubrication will not be permitted.

3.4.4 Threads -

3.4.4.1 Class 1 rod ends - Threads shall be rolled before or after heat treatment and conform to MIL-S-8879. Opitonal thread runout shall be in accordance with MIL-B-7838.

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3.4.4.2 Class 2 rod ends - Threads shall conform to MIL-S-8879.

3.5 Performance -

3.5.1 Ultimate static load - No fracture of the bearing shall occur when the ultimate static load specified on the applicable rod end MS is applied in accordance with 4.6.1.

3.5.2 Axial static proof load - No pushout of the bearing cartridge from the rod end body shall occur when the axial proof load on the applicable MS is applied as specified in 4.6.2.

3.5.3 Fatigue load - The rod end bearings shall be capable of withstanding a minimum of 50,000 cycles of loading when tested in accordance with 4.6.3.

3.5.4 Self-alignment - The bearing shall be self-aligning and permit the angular displacement specified on the applicable rod end MS.

3.5.5 No-load rotational breakaway torque - When tested in accordance with 4.6.4, the no-load breakaway torque shall be within the limits of the values specified on the applicable rod end MS.

3.6 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's part numbers.

3.7 Identification of product - Each bearing shall be marked with the military identifying part number and the manufacturer's name or trademark, and manufacturer's part number prefixed by (MFR). Where practicable, identification shall appear on the side face of the rod end body; otherwise identification shall appear on the periphery of the rod end body. Metal impression stamping shall not be permitted.

3.8 Workmanship - The bearing shall be free of toolmarks, chatter waves, rust, grinding scratches, pits, or any other defects that may adversely affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Qualification test records - The manufacturer shall maintain a record showing quantitative results for all tests required by this specification. This record shall be available to the purchaser and shall be signed by an authorized representative of the manufacturer or the testing laboratory, as applicable.

4.2 Classification of inspection - The examination and testing of the bearings shall be classified as:

(a) Qualification inspection (4.3)
(b) Quality conformance inspection (4.4)

4.3 Qualification inspections -

4.3.1 Sampling instructions - Qualification inspection samples shall consist of 10 bearings of the - 6, - 8, and -12 sizes of each class (1 and 2), for which qualification is desired. All bearings necessary for tests specified herein shall be furnished by the manufacturer. When approved, the - 6, - 8, and -12 sizes of a class will qualify the - 3 through -16 sizes of that class. Samples shall be identified as required and forwarded to the activity designated in the letter of authorization from the activity responsible for qualification (see 6.3).

4.3.2 Certified test report - The manufacturer shall furnish a certified test report showing that the manufacturer's product satisfactorily conforms to this specification. The test report shall include, as a minimum, actual results of the tests specified herein. When the report is submitted, it shall be accompanied by a dated drawing which completely describes the manufacturer's product by specifying all dimensions and tolerances, composition of the ball and outer race material, lubricant if used, coating or plating, and heat treatment. The manufacturer's part number for each size shall be included on the drawing.

4.3.3 Inspections - Qualification inspections shall include all the examinations and tests of this specification. The minimum number of samples per test shall be in accordance with table I.

TABLE I. Qualification inspection samples.

Examination and test	Paragraph No.	Samples to be tested
Examination of product	4.5.1	5
Preparation for delivery	4.5.2	5
Ultimate static load	4.6.1	3
Axial static proof load	4.6.2	1/ 3
Fatigue load	4.6.3	3

1/ Qualification of class 1 will qualify class 1 and class 2.

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4.4 Quality conformance inspections - The quality conformance inspections of the bearings shall consist of the following examinations and tests to determine conformance of the bearings to the requirements of this specification and the applicable MS with regard to:

(a) Dimensions	(3.4.1)	(4.5.1)
(b) Identification of product	(3.7)	(4.5.1)
(c) Workmanship	(3.8)	(4.5.1)
(d) Preparation for delivery		(4.5.2)
(e) No-load breakaway torque		(4.6.4)
(f) Hardness		(4.6.5)
(g) Magnetic particle inspection		(4.6.6)

4.4.1 Inspections - The bearing manufacturer shall be responsible for accomplishing the quality conformance inspections specified herein.

4.4.2 Lot - The lot definition, formation, and size shall be in accordance with MIL-STD-105.

4.4.3 Sampling - The sample bearings shall be selected in accordance with MIL-STD-105, inspection level II, acceptable quality level of 1.0 percent, except that inspection for no-load breakaway torque and magnetic or penetrant inspection of the rod end body shall be 100 percent.

4.5 Examinations -

4.5.1 Examination of product - The bearings shall be examined to determine conformance to the requirements of this specification and the applicable MS for material, plating, dimensions, finish, identification of product, workmanship, and requirements not covered by tests.

4.5.2 Inspection of preparation for delivery - Preservation, packaging, packing, and marking shall be inspected to determine conformance to section 5.

4.6 Test Methods - Unless otherwise specified, all tests shall be conducted at room temperature.

4.6.1 Ultimate static load - The bearing shall be installed in a test fixture as shown on figure 1, using a 0.0000 to 0.0015-inch loose fit with the shaft. The appropriate ultimate static load specified on the rod end MS shall be applied.

4.6.2 Axial static proof load - The bearing shall be mounted on a rigid fixture as shown on figure 2, so that only the rod end body is supported. The axial static proof load shall be applied to the ball and maintained by the bearing for 30 seconds.

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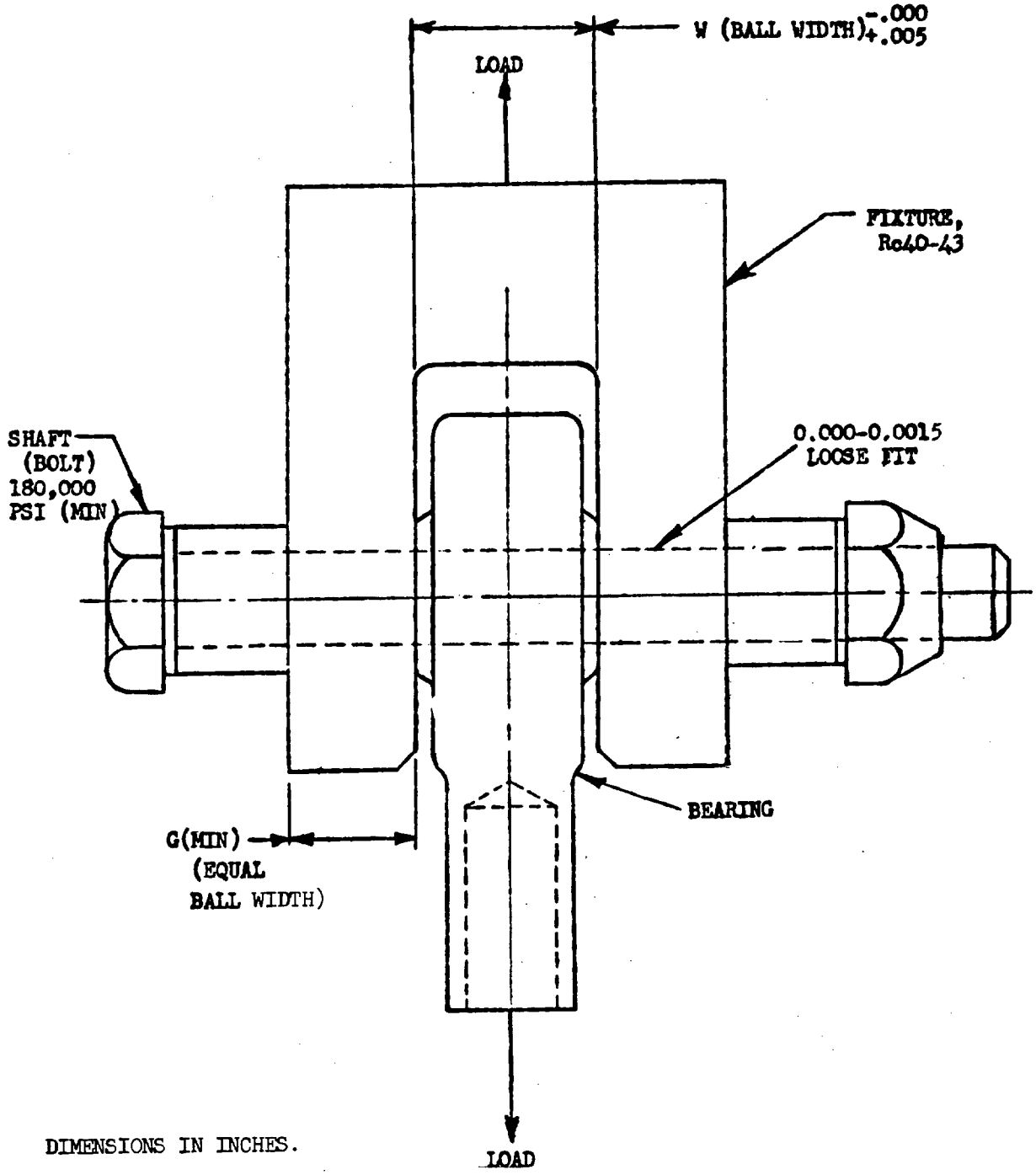


FIGURE 1. Ultimate and fatigue test fixture.

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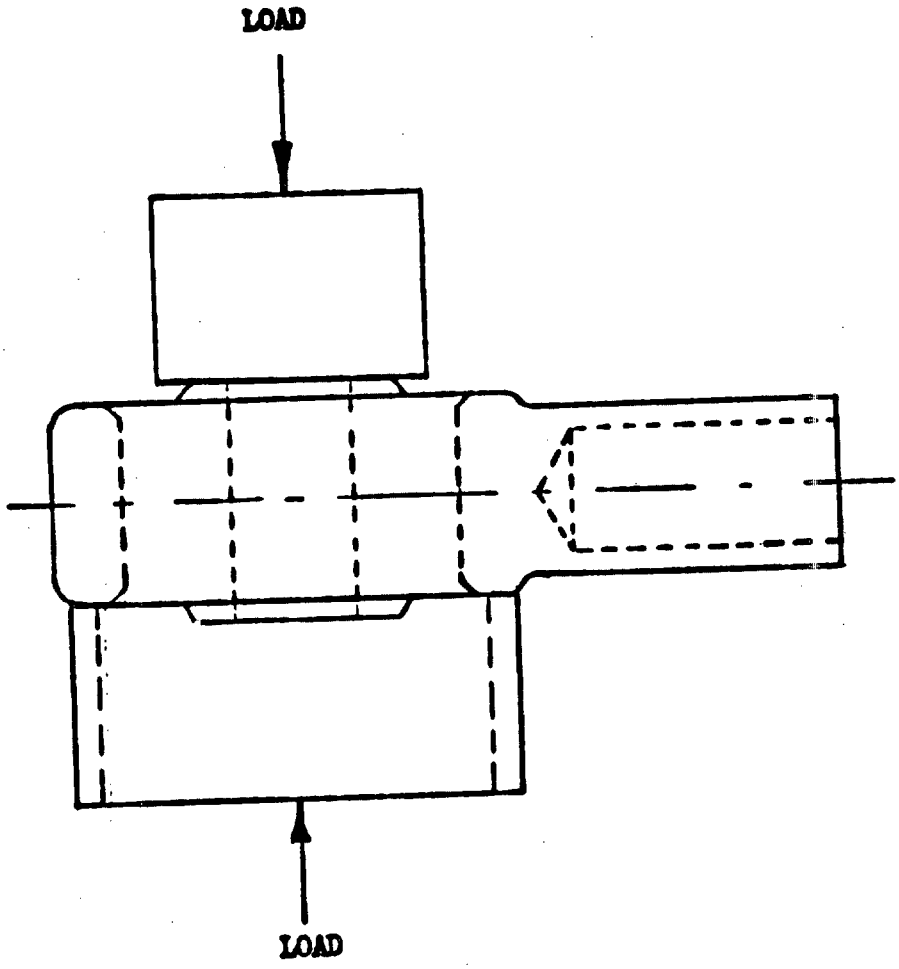


FIGURE 2. Axial load test setup

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4.6.3 Fatigue load - The bearing shall be installed in a fatigue machine using a test fixture as shown on figure 1, with a 0.0000 to 0.0015-inch loose fit with the shaft. The loading shall be tension-tension with the maximum load equal to the fatigue load on the applicable MS and the minimum load equal to 10 percent of the fatigue load. Loads shall be maintained within ±2 percent. Testing speeds shall not exceed 2,800 cycles per minute.

4.6.4 No-load rotational breakaway torque - The no-load rotational breakaway torque shall be determined by holding the outer race of the bearing fixed while rotating the ball about the bearing axis. The outer race shall be held in such manner as to minimize bearing distortion and the resultant effect on the bearing preload. Torque shall be applied gradually to the ball; the minimum torque required to start the ball moving shall be recorded. The no-load rotational breakaway torque shall be as specified on the applicable MS. The ball shall be hand rotated through several revolutions immediately prior to testing.

4.6.5 Hardness - The bearings shall be tested for hardness using a standard Rockwell hardness tester. Readings for the rod end body shall be taken on the flat at the head shank junction.

4.6.6 Magnetic particle inspection - Alloy steel rod end bodies shall be inspected in accordance with MIL-I-6868. The inspection shall be by both the longitudinal and circular methods. Certification of such inspection shall be furnished with qualification samples and quality conformance tests.

5. PREPARATION FOR DELIVERY

5.1 Cleaning, preservation, and packaging -

5.1.1 Level A - Cleaning, preservation, and packaging shall be in accordance with level A of MIL-B-197.

5.1.2 Level C - Cleaning, preservation, and packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing -

5.2.1 Level A - Bearings, cleaned, preserved, and packaged as specified in 5.1.1, shall be packed for overseas shipment in accordance with MIL-B-197.

5.2.2 Level B - Bearings, cleaned, preserved, and packaged as specified in 5.1.1 shall be packed for domestic shipment and storage in accordance with MIL-B-197.

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5.2.3 Level C - Bearings, cleaned, preserved, and packaged as specified in 5.1.2, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. The containers shall be in accordance with the Uniform Freight Classification Rules or regulations of other carriers, as applicable to the mode of transportation.

5.3 Marking of shipments - Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall include:

BEARINGS, PLAIN, ROD END, TFE LINED, SELF-ALIGNING

6. NOTES

6.1 Intended use - These bearings are intended primarily for use in airframe applications of highloads at low rotational oscillatory speeds. For specific design information on the capability of these bearings under particular load, speed and wear/life conditions the user is referred to The Airframe Design Guide.

6.2 Ordering data - Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) MS part number (see 3.3).
(c) Class of bearing (see 1.2).
(d) Applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

6.3 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Navy Department, Washington, D.C. 20360; however, information pertaining to qualification of products may be obtained from the Naval Air Development Center, AMD, Code MAEM, Warminster, Pennsylvania 18974.

6.3.1 Qualification tests will be authorized only upon presentation of certified test reports and drawings indicating that the bearings conform to this specification.

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Air Force - 11
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Reviewer activities:
Air Force - 84
DSA - IS

Preparing activity:
Navy - AS
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE MIL-B-8948A BEARINGS, PLAIN, ROD END, TFE LINED, SELF-ALIGNING

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO (If "Yes", in what way?)

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